

## Short Communication

## A test of island plant syndromes using resource-use traits

Andrea C. Westerband<sup>1†\*</sup>, Tiffany M. Knight<sup>2,3,4</sup>, and Kasey E. Barton<sup>1</sup><sup>1</sup>School of Life Sciences, University of Hawaii at Manoa, 3190 Maile Way, Room 101, Honolulu 96822, HI, USA<sup>2</sup>Department of Community Ecology, Helmholtz Centre for Environmental Research-UFZ, Halle (Saale), Germany<sup>3</sup>Institute of Biology, Martin Luther University Halle-Wittenberg, Halle (Saale), Germany<sup>4</sup>German Centre for Integrative Biodiversity Research (iDiv) Halle-Jena-Leipzig, Leipzig, Germany<sup>†</sup>Current address: Andrea C. Westerband, Hawkesbury Institute for the Environment, Western Sydney University, Richmond, NSW 2753, Australia

\*Author for correspondence. E-mail: andreawesterband@gmail.com

Received 24 January 2023; Accepted 6 October 2023

**Abstract** Despite representing a fraction of the global terrestrial surface area, oceanic islands are disproportionately diverse in species, resulting from high rates of endemism. Island plants are thought to share a unique phenotype—referred to as an island syndrome—which is thought to be driven by convergent evolution in response to selection by shared abiotic and biotic factors. One aspect of the island plant syndrome that has received relatively little research focus is that island plants are expected to have converged on conservative resource use associated with slow growth rates and weak competitive abilities. Here we tested whether native, woody Hawaiian plant species are phenotypically distinct—with more resource-conservative leaf traits—compared to a globally distributed sample of continental species. Using an archipelago-wide trait data set, we detected that on average, native Hawaiian species had lower leaf nutrient concentrations overall, and lower nutrient concentrations at high leaf mass per area, but no other phenotypic differences compared with continental plants. There was also considerable overlap in the trait spaces of native Hawaiian species and continental species. Our findings indicate that an island plant syndrome for leaf traits is not present in the Hawaiian flora, and that island species can demonstrate extensive variation in their resource-use strategies, on a scale that is comparable with that of continental species worldwide.

**Key words:** endemic species, functional traits, Hawaii, island syndromes, leaf economics spectrum, phylogenetic disharmony.

## 1 Introduction

Oceanic islands harbor unique biodiversity resulting from long-term isolation, dispersal limitations, and ongoing adaptive radiations (Carr, 1987; Wagner et al., 1999; Price & Wagner, 2018). The former is thought to lead to absent guilds or taxa (Price & Wagner, 2018) while the latter drives high diversity in a subset of lineages (Carr, 1987; Wagner et al., 1999). Together, these mechanisms underlie the phylogenetic disharmony between islands and continental ecosystems. In addition, island biotas are predicted to be phenotypically unique, as a consequence of phylogenetic disharmony, environmental filtering, and convergent evolution on islands (Carr, 1987; Simberloff, 1995; Gillespie & Baldwin, 2010; Burns, 2019). For plants, this gives rise to the predicted “island plant syndromes” including a high prevalence of woody species, loss of seed dispersibility, generalized floral anatomy, and weak defenses against natural enemies (Carlquist, 1974; Kores, 1980; Bowen &

Van Vuren, 1997; Ziegler, 2002; Funk & Throop, 2010; Wagner et al., 2017). It has also been predicted that island plants have evolved conservative resource use strategies associated with slow growth as a consequence of relaxed competition in low-diversity island plant communities (Simberloff, 1995; Denslow, 2003).

Island plant syndrome predictions are typically investigated by comparing the traits of native island species against nonnative continental plants growing *in situ* (Mologni et al., 2022) or by comparing a single species growing on a continent to the same species on an island (Burns, 2016). For example, a number of previous studies have tested whether natives and invasives on islands differed in their stature, dispersal mode, and growth rates (Heberling & Mason, 2018; Mathakutha et al., 2019; Sperry et al., 2021) because these traits are thought to be associated with competitive ability and persistence. However, comparisons using nonnative invasives may not be suitable given that invasive plants are not a random subset of continental plants. Indeed, a number

This is an open access article under the terms of the Creative Commons Attribution-NonCommercial-NoDerivs License, which permits use and distribution in any medium, provided the original work is properly cited, the use is non-commercial and no modifications or adaptations are made.

of studies have shown that plant species that become invasive outside of their native range are characterized by the presence of more resource-acquisitive leaf traits (e.g., higher leaf nutrient concentrations, higher rates of photosynthesis) than noninvasive continental species (Peñuelas et al., 2010; van Kleunen et al., 2010; Montesinos, 2022). This patterning can potentially exacerbate differences between native island species and invasive continental species. To more precisely test whether the traits of island plants differ from those of continental plants, a comparison of continental plants sampled on continents versus island plants sampled on islands is needed (Barton & Fortunel, 2023) and is the subject of the current study.

We tested the prediction that Hawaiian species have an island “leaf economics” syndrome using a leaf trait data set spanning four of the major Hawaiian Islands. Namely, we predicted that natives would have traits aligned with a conservative resource-use strategy when plotted against a global sample composed of continental species. Based on previous studies (Pattison et al., 1998; Baruch & Goldstein, 1999; Santiago et al., 2000; Stratton & Goldstein, 2001; Cordell et al., 2002; Funk, 2013; Cavaleri et al., 2014), we also predicted that Hawaiian natives would have lower leaf nitrogen (N) and leaf phosphorus (P) concentrations, lower photosynthesis ( $A_{\text{mass}}$ ), and higher leaf mass per area (LMA) than the global continental sample, and stronger clustering toward the “slow” end of the leaf economics spectrum (LES) (Wright et al., 2004). The Hawaiian flora presents an exemplary case of phylogenetic disharmony (König et al., 2021) resulting from its extreme isolation and repeated adaptive radiations (Price & Wagner, 2004). Despite this disharmony, we predicted that there would be relatively strong effects of phylogenetic relatedness on the above trait comparisons. In other words, we anticipated that by controlling for phylogenetic relatedness in our analyses, that the evidence for an island syndrome in leaf traits would be stronger. We focus on bivariate and multivariate trait relationships, which can reveal ecologically meaningful variations in resource use strategies that are not apparent from examining individual traits (Warton et al., 2006). This study builds on a previous study by Heberling & Fridley (2012), which found that when compared with a sample of continental species from tropical regions, Hawaiian species did not differ in LES trait covariation patterns ( $A_{\text{mass}}-N_{\text{mass}}$  in that study). To our knowledge, ours is the first study to test for evidence of island syndromes using leaf economic traits and thus contributes to a growing area of research known as functional island biogeography (Ottaviani et al., 2020; Schrader et al., 2021).

## 2 Material and Methods

### 2.1 Trait data

The Hawaiian trait data set used in this study is described in detail in Westerband et al. (2020), but briefly, we sampled 52 native, woody Hawaiian species (Table S1) across 21 sites on four Hawaiian Islands: Hawaii, Maui, Oahu, and Kauai, from August 2016 to November 2017. Replication within species varied across sites and in total, we sampled 560 individuals. We sampled a mix of dominant and rare species, where the dominant species were chosen based on their occurrence

with *Metrosideros polymorpha* (Craven et al., 2018), a dominant keystone species across the archipelago (Barton et al., 2021). Indeed, many of the focal species included herein are among the 25 most dominant species in Hawaii (Craven et al., 2018). There was also variation in growth form within our sample, including shrubs, trees, and climbers (Table S1). Twenty-one of the 52 species (40%) are trees, and the remaining species are shrubs or climbers.

Rare species were included alongside dominants, as rare species have been shown to have a disproportionate impact on functional richness in species assemblages (Leitão et al., 2016), yet are largely ignored in functional trait studies. The inclusion of rare and common species resulted in uneven replication across species. Therefore, the sampling effort reflects species' abundances across the study sites. An alternative strategy would be to use community-level data to choose species based on their relative abundances at each site. However, community-level data was not available for our sites, and characterizing each community would preclude a geographically widespread sampling scheme in favor of more detailed, localized data for each site. Notably, the breadth of the sampling in the present study far surpasses those of previous studies (see Section 1) in terms of the number of sites (>20) and islands (four) included. Conversely, the global and regional data sets included herein are more likely to include common species alone (see Section 2.2).

Sampled plants span broad abiotic gradients: sea level—2000 m asl., 400–5600 mm in mean annual precipitation, and 10–24 °C in mean annual air temperature. For each individual plant, we collected 1–20 leaves per plant but typically five, which were fully mature, sun-exposed, and at the terminal ends of the branches as per Perez-Harguindeguy et al. (2016). This sampling scheme reduced intraindividual variation. The number of leaves sampled per species depended on their size because only 1 g of leaf tissue was needed for the nutrient analyses (see below). We measured four leaf traits associated with the leaf economics spectrum, or LES (Wright et al., 2004). These include LMA ( $\text{g/m}^2$ ), leaf mass-based light-saturated rate of photosynthesis ( $A_{\text{mass}}$ ,  $\text{nmol CO}_2/\text{g per s}$ ), leaf nitrogen (N) concentration (mass-based, %N), and leaf phosphorus (P) concentration (mass-based, %P).

### 2.2 Data compilation

Hawaii's disharmonic flora makes it challenging to identify the most robust continental flora for comparison. Native Hawaiian species have been shown to originate from both tropical and temperate regions, and the origins of approximately half the flora remain unknown (Price & Wagner, 2018). For these reasons, we extracted trait data for continental species growing in both tropical and temperate regions. We excluded any data collected on islands globally as well as any native Hawaiian species sampled on continents (Tables S2, S3). It is possible, but unlikely, that the continental data set includes non-Hawaiian island species sampled on continents. The data were extracted from five publicly available data sets (two continental and three global, Table S3) via the online data repository, TRY (Kattge et al., 2020). The selection of continental species was haphazard in that we did not purposefully select closely related species or species inhabiting a particular geographic region or with a specific

origin. We also extracted growth form for all the species in our data set by cross-referencing our species with the “Global Spectrum of Plant Form and Function Dataset” (Díaz et al., 2022) or via a Google search. The final data set included 906 species: 52 native Hawaiian species from the present study and 854 non-Hawaiian continental species (Figs. S1, S2). Of the continental species (Table S2), most (54.5%) were trees, followed by shrubs or shrub-like trees (27.8%). A minor percentage (14.65) were nonwoody, including grasses, herbs, and ferns. It is likely that these species are primarily common (or locally dominant species), however, information regarding local species abundances was not available for the continental and global data sets. Only species that contained data for all four focal traits were included. Species names were cross-referenced with their accepted taxonomic names according to The Plant List (V1.1; <http://www.theplantlist.org/>). A total of 36 species were pruned from the final phylogeny (Fig. S2) because their names were not identified beyond genus level or could not be placed on the reference phylogenetic tree from the Open Tree of Life project v 9.1 (GBOTB) (Smith & Brown, 2018). Prior to our analyses, all traits were  $\log_{10}$ -transformed to improve the normality of the residuals, and the transformed data were used to calculate species-site means for all data sets.

### 2.3 Statistical analysis

All statistical analyses were carried out using the statistical software program R v. 3.6.1 (R Core Team, 2021). We tested for a phylogenetic signal in the leaf traits, noting that several of the Hawaiian genera have undergone extensive adaptive radiations, which could result in strong phylogenetic signaling (Gardner, 1976; Carlquist, 1980; Carr & Kyhos, 1981; Givnish et al., 2009). A strong phylogenetic signal would result from closely related species having more similar traits than species drawn at random from the phylogenetic tree (Blomberg & Garland, 2002). We calculated Blomberg's  $K$  and Pagel's  $\lambda$ , which compare a given phylogeny to one expected under a Brownian model of evolution (Münkemüller et al., 2012). Values close to zero indicate the lack of phylogenetic signal. Pagel's  $\lambda$  is essentially a scaling parameter that transforms the given phylogeny to fit a phylogeny that would be expected under a Brownian model of evolution (Münkemüller et al., 2012). In contrast, Blomberg's  $K$  is a function of two ratios, and it compares the variation in the observed data to that expected under Brownian motion (Blomberg et al., 2003; Revell & Harmon, 2008). We tested for a phylogenetic signal using the “phylosig” function from the phytools package (Revell, 2012).

We used a phylogenetically controlled analysis of variance (pANOVA) to test whether the trait means for continental species differed from those of native Hawaiian species while accounting for nonindependence in the data points due to phylogenetic relatedness (Revell, 2012). The pANOVA was based on Garland et al. (1993) where a null distribution of  $F$ -ratios is derived from the simulated evolution of continuous traits (assuming Brownian motion) along the branches of a supplied phylogenetic tree, without regard to the group membership of the species at the tips of the phylogeny. We used 1000 simulations in the present study.

To test whether traits associated with the LES covaried in different ways for the Hawaiian and continental species, we tested for homogeneity in the slopes of bivariate trait relationships using standardized major axis regression (SMA). Significant deviations in the slopes would indicate distinct resource use strategies between native Hawaiian species and the continental species. In this analysis, we allowed the intercepts to vary rather than setting them to zero, and we only tested for differences in slopes and not elevation (i.e., intercepts) (Warton et al., 2006, 2012). We then used the full data set to test whether phylogenetic relatedness influenced the bivariate trait relationships, and this was implemented using a phylogenetic-reduced major axis regression (pRMA). In this analysis, it was not possible to test for group differences and/or to calculate slopes for each group. Instead, this analysis tested whether phylogenetic relatedness influenced the strength and direction of the trait relationships, where the null hypotheses varied with the trait relationship. We repeated this analysis, once using the full data set and then again without herbaceous species ( $N = 728$ ). The SMA was carried out using the smatr package (Warton et al., 2012) while the pANOVA and pRMA were carried out using the phytools package (Revell, 2012).

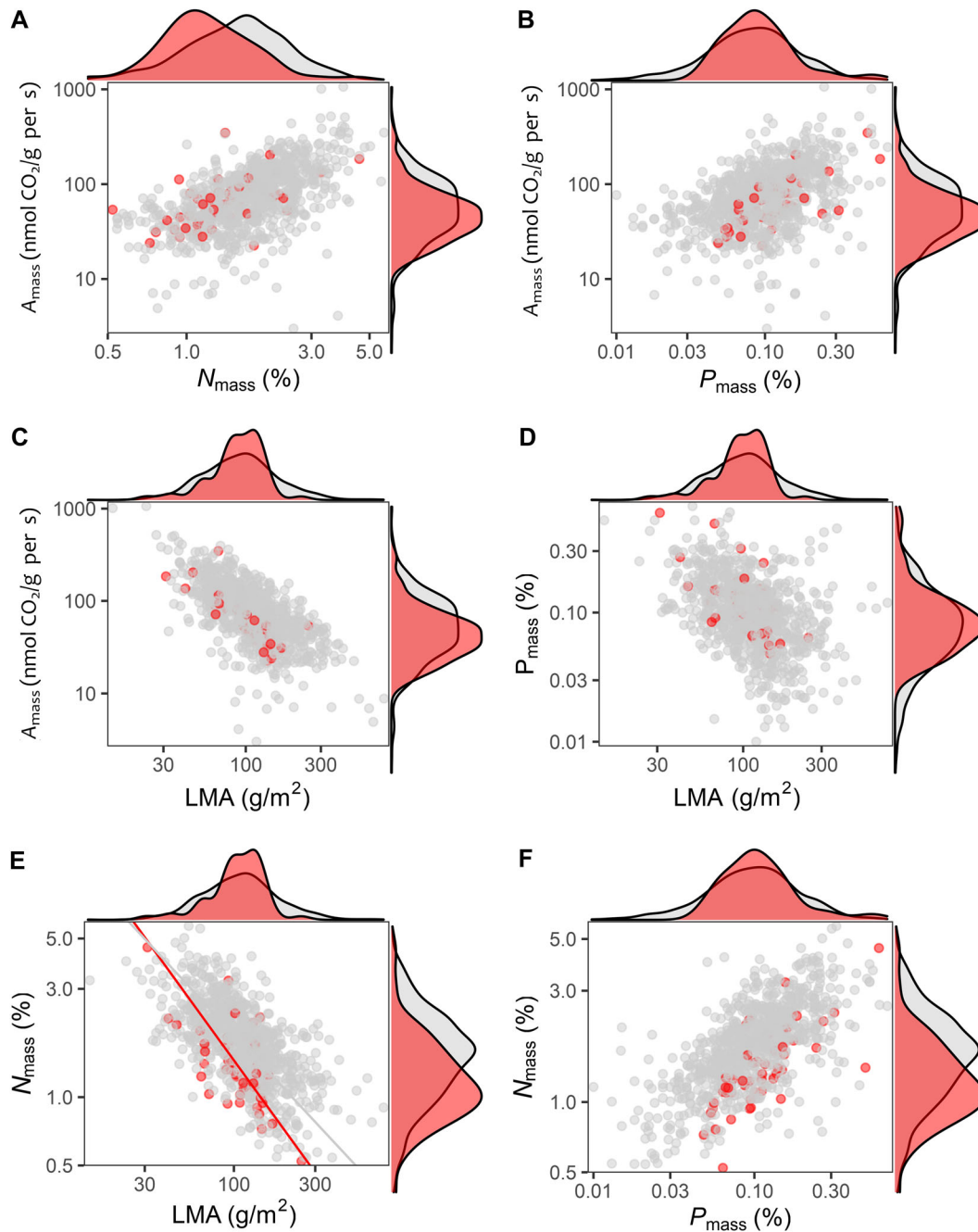
To test whether native Hawaiian plants and the continental species occupied different regions of the trait space, we carried out a phylogenetically controlled principal components analysis (pPCA) (Revell, 2009, 2012) using all four LES traits. We conducted this analysis by including woody and herbaceous species and then including woody species alone. We visually assessed the extent of overlap. Principal component loadings were used to determine which traits were responsible for the separation in trait space. We also carried out a phylogenetic PCA on the Hawaiian species alone. The pPCAs were carried out using the “phyl.pca” function in the phytools package (Revell, 2009, 2012).

## 3 Results

There was evidence that closely related species had more similar trait values than expected by chance, whether we used the full data set or native Hawaiian species alone (Table S4). The evidence was stronger when evaluated on the basis of Pagel's  $\lambda$ , whereas Blomberg's  $K$  only detected a phylogenetic signal in  $N_{\text{mass}}$  and  $P_{\text{mass}}$  within the Hawaiian species but not in the full data set that included continental and Hawaiian species.

Relative to continental species, native Hawaiian species had significantly lower  $N_{\text{mass}}$  ( $F = 16.7$ ,  $P = 0.01$ ) but did not differ in LMA,  $P_{\text{mass}}$  or  $A_{\text{mass}}$  (pANOVA,  $P > 0.05$ ) (Table S5). On average,  $N_{\text{mass}}$  for native Hawaiian species was 1.49% compared with 1.89% in the continental species, a difference of approximately 27%.

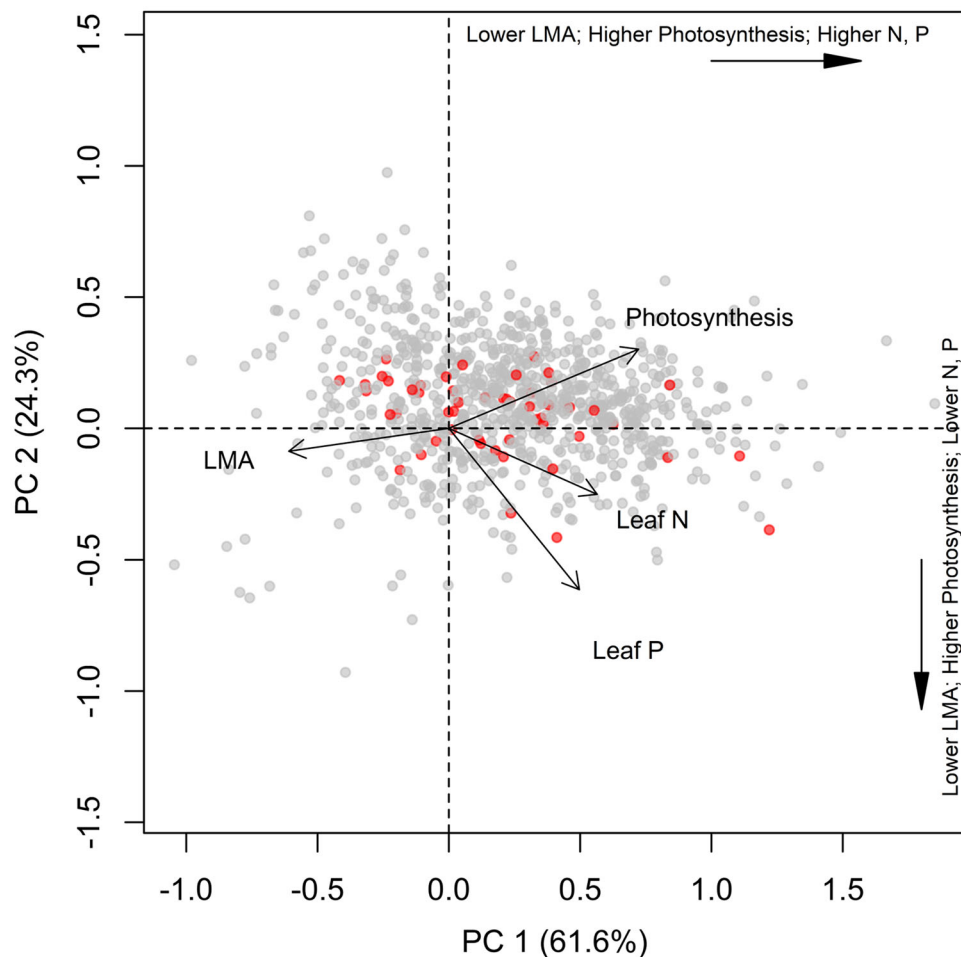
We evaluated whether phylogenetic relatedness influenced bivariate trait relationships using a phylogenetic tree that included both native Hawaiian species and the globally distributed continental species (Fig. 1). We found that accounting for phylogeny resulted in flatter-than-expected relationships between LMA and each of  $A_{\text{mass}}$  and  $P_{\text{mass}}$  but a steeper relationship between LMA and  $N_{\text{mass}}$  (Table S6). There was no effect on the remaining traits. Based on this



**Fig. 1.** Variation in leaf traits of native Hawaiian species (red) and continental species (gray). Relationships between mass-based photosynthesis and (A) leaf nitrogen or (B) leaf phosphorus. Relationships between leaf mass per area (LMA) and each of (C) mass-based photosynthesis, (D) leaf phosphorus, and (E) leaf nitrogen. Relationship between leaf nitrogen and phosphorus, on a mass basis (F). Each point represents a species mean and only significant relationships are shown, as indicated with regression lines. The center portion of each figure shows a scatterplot, where axes have been logarithmically transformed. Lines shown in panel (E) indicate slopes from standardized major axis regression, which does not account for phylogenetic relatedness. Density plots are shown along the margins of each scatterplot. See Supplementary information for phylogenetically reduced major axis regression and for the same analysis without herbaceous species.

analysis, we tested for slope differences between the native Hawaiian species and the continental species and found a more steeply negative relationship between LMA and  $N_{\text{mass}}$  in the native Hawaiian species compared with the continental

species ( $P = 0.03$ , slope of  $-1.04$  versus  $-0.80$ , respectively) (Fig. 1; Table S7); native Hawaiian species had lower leaf nitrogen concentrations than continental species when LMA was high. There were no other differences between the



**Fig. 2.** Trait space of native Hawaiian species (red) superimposed onto a globally distributed sample of continental species (gray). Each point represents a species. Species clustered toward the lefthand side of the trait space have higher leaf mass per area (LMA), lower leaf N and leaf P, and lower rates of photosynthesis, that is, a more conservative resource-use strategy. Individuals clustered toward the lower portion of the trait space have lower LMA, higher photosynthesis, lower leaf N and lower leaf P, that is, a combination of resource-conservative and -acquisitive traits. See Supplementary information for the same analysis without herbaceous species.

native Hawaiian species and the continental species, which was contrary to the expectation that native Hawaiian species should have traits that reflect conservative resource use. We also evaluated the slope differences between native woody Hawaiian species and woody continental species and found no difference in the slope relationships (Table S8) but lower  $R^2$  values, indicating weaker explanatory power.

From the pPCA, there was little evidence that native Hawaiian species occur on the more conservative end of the LES (Fig. 2), which was captured on the first axis (Table S9). An additional pPCA on woody species alone produced similar findings (Fig. S4), albeit with a slight increase in the explanatory power of the second axis. Indeed, native Hawaiian species were spread evenly across the trait space. The first and second axes explained 62% and 24% of the trait variation, respectively. A phylogenetically controlled PCA using only the native Hawaiian species (Fig. S3; Table S10) was nearly identical to the pPCA on the full data set.

## 4 Discussion

We predicted that native Hawaiian species would exhibit a highly conservative resource-use strategy compared with a globally distributed sample of continental species, in what is, to the best of our knowledge, the first island versus continental test of island syndromes using leaf economic traits. Using a combination of univariate, bivariate, and multivariate analyses to study differences in traits and trait covariation, we found little evidence that native Hawaiian species are fundamentally different from continental species or that they exhibit an island syndrome in their leaf traits. Compared with the continental species, there was less leaf N overall and less leaf N at high LMA (Fig. 1), which is indicative of more conservative resource use. Leaves with low LMA are typically associated with faster whole-plant growth as these leaves are relatively cheaper to produce than high LMA leaves (Reich, 2014). Interestingly, there were two native Hawaiian species that were positioned toward the more

acquisitive end of the LES: *Erythrina sandwicensis* (Fabaceae) and *Pisonia wagneriana* (Nyctaginaceae). *Erythrina sandwicensis* is one of Hawai'i's few deciduous native trees, dropping its leaves in the summer presumably for the conservation of water. This species is known to occupy dry forests on the leeward slopes of the major Hawaiian Islands, with an elevation of 0 to ca. 600 m (Wagner et al., 1999). Less is published about *P. wagneriana*, which is restricted to the island of Kauai and is categorized as rare. Based on the considerable overlap in trait space between native Hawaiian species and the global sample, we thus reject the claim that Hawaiian plants are phenotypically distinct from a global sample of continental species in their resource use strategies (Leishman et al., 2007). Similarly, Heberling & Fridley (2012) found that when compared with a sample of continental species from tropical regions, Hawaiian species did not differ in LES trait covariation patterns ( $A_{\text{mass}}-N_{\text{mass}}$  in that study).

If LES traits associated with more vigorous life histories are present in native Hawaiian species, it suggests that these resource-use strategies are present in at least some island species, likely those inhabiting high-resource environments. Indeed, Westerband et al. (2020) found that under a subset of climates, native Hawaiian species had more resource-acquisitive traits than co-occurring nonnative invaders, which may explain the overlap detected herein. We further note that plant functional groups (i.e., growth form) likely contribute to the extensive trait overlap between native Hawaiian species and the continental sample (Tables S1, S2). For example, large-leaved Hawaiian species (e.g., *Pandanus tectorius*) and pioneer species capable of establishing on recently formed volcanic soils (e.g., *Metrosideros polymorpha*) are likely to possess more resource acquisitive traits than species restricted to the forest understory (e.g., species in the genus *Coprosma*). Similarly, herbaceous species are typically more resource-acquisitive than woody species (Díaz et al., 2022) and were more abundant in the continental data sets. While it is possible that including herbaceous species alongside woody species in the continental data set could have exaggerated differences between native Hawaiian species and continental species, there was no evidence of this in the present study (Fig. S4; Table S8). Our findings indicate that the overlap in trait space was not driven by the presence of herbaceous species on continents.

The lack of support for an island plant syndrome in leaf traits, as evidenced by conservative resource use (Simberloff, 1995; Denslow, 2003), suggests that the functional strategies of Hawaiian plants are more diverse than expected. Studies characterizing other facets of island plant functional strategies would shed light on the extent of functional diversity within island floras (Schrader et al., 2021). For example, drought tolerance traits (e.g., seed dormancy, rooting depth) or traits associated with light capture (e.g., height) may also be under strong selection in native Hawaiian forests (Kagawa et al., 2009; Biddick et al., 2019), which have relatively open canopies compared with many other tropical forests (Barton et al., 2021). Because of the ongoing and increasingly widespread effects of anthropogenic global change, including habitat alteration and fragmentation, species introductions, and climate change, disturbance regimes are intensifying on islands (Caujapé-Castells et al., 2010; Timm et al., 2015; Trauernicht et al., 2015;

Frazier & Giambelluca, 2017), which is likely to alter resource availability and thus, impose strong selection on island plant functional strategies. Future work exploring the role of disturbance on island plant functional diversity is needed to characterize these relationships and project island plant resilience under future change (Murakami et al., 2013).

Island plant syndromes are thought to have evolved simultaneously (i.e., convergent evolution) due to similarities in biotic and abiotic filters. This likely reflects the considerable environmental heterogeneity that occurs within volcanic islands with steep elevational gradients such as the Hawaiian Islands. To further test whether island species are typically more resource-conservative, future studies should examine whether the patterns we detected for Hawaiian plants hold on other island archipelagos, including those in temperature and tropical climates. Considering the extreme isolation and endemism of the Hawaiian flora, it is feasible that Hawaiian species deviate from a pattern shared among other island plants. Furthermore, given the extensive adaptive radiations that characterize the Hawaiian flora (e.g., the lobeliads) we surmise that closely related species occupying distinct microenvironments may contribute to the wide range of variation in trait space (Fig. 2) and to greater levels of resource acquisitiveness than would otherwise be expected. Indeed, a previous study found evidence of adaptive radiations in the photosynthetic physiology of 11 lobeliad species spanning a wide range of light environments in the field (Givnish & Montgomery, 2014). The authors determined that species that typically occupied brighter environments had higher  $A_{\text{mass}}$  when reared in high levels of ambient light than those species that typically occupied shadier environments (Givnish & Montgomery, 2014). Thus, future studies should compare the resource use strategies—using trait means but also trait distributions (Hulme & Bernard-Verdier, 2018)—of continental species to highly diverse taxonomic groups in Hawaii that are known to have undergone adaptive radiations including the lobeliads (Campanulaceae) but also the melicopes (Rutaceae) (Stone, 1966; Wagner et al., 1999), silverswords (Asteraceae) (Baldwin & Sanderson, 1998), and the genus *Schiedea* (Caryophyllaceae) (Wagner et al., 1995).

The findings from the present study clarify that in highly isolated ecosystems, such as the Hawaiian Islands, leaf traits (specifically those associated with carbon economy) are unlikely to have evolved in tandem to produce a phenotypically distinct island-plant strategy. It is possible that selection on other traits or trait networks (e.g., coordinated water transport mechanisms) are under stronger selection than carbon economy traits, and these should be examined more closely. Ecologically, the consequences of long-term isolation should be the same on continental and oceanic islands, which provides ongoing opportunities to study eco-evolutionary hypotheses regarding mechanisms of speciation and diversification. Indeed, isolated habitats on continents, such as mountaintops and valleys, may also be functionally considered “islands” as they have conditions that impose strong selection on colonizers (Ackerly, 2003). Such studies will surely increase our fundamental understanding of plant trait evolution in isolation and are particularly relevant in light of ongoing anthropogenic disturbance, urbanization, and habitat fragmentation worldwide.



## Acknowledgements

The authors thank the following graduate students who assisted with the data collection: T. Lum and K. Bogner. Additional help was provided by undergraduate students: A. Wong, A. Shiarella, and C. Nomura. A.C.W. led the data collection and conducted the analyses and writing with assistance from K.E.B. and T.M.K. Funding was provided by the University of Hawaii at Manoa College of Natural Sciences to K.E.B., by the Helmholtz Association as part of the Helmholtz recruitment initiative to T.M.K. Open access publishing facilitated by Western Sydney University, as part of the Wiley - Western Sydney University agreement via the Council of Australian University Librarians.

## Data Availability Statement

The data used in this study will be made available upon request by the corresponding author.

## References

- Ackerly DD. 2003. Community assembly, niche conservatism, and adaptive evolution in changing environments. *International Journal of Plant Sciences* 164: S165–S184.
- Baldwin BG, Sanderson MJ. 1998. Age and rate of diversification of the Hawaiian silversword alliance (Compositae). *Proceedings of the National Academy of Sciences of the United States of America* 95: 9402–9406.
- Barton KE, Fortuon C. 2023. Island plant functional syndromes and competition with invasive species. *Journal of Biogeography* 50: 641–653.
- Barton KE, Westerband A, Ostertag R, Stacy E, Winter K, Drake DR, Fortini LB, Litton CM, Cordell S, Krushelnycky P. 2021. Hawai'i forest review: Synthesizing the ecology, evolution, and conservation of a model system. *Perspectives in Plant Ecology, Evolution and Systematics* 52: 125631.
- Baruch Z, Goldstein G. 1999. Leaf construction cost, nutrient concentration, and net CO<sub>2</sub> assimilation of native and invasive species in Hawaii. *Oecologia* 121: 183–192.
- Biddick M, Hendriks A, Burns K. 2019. Plants obey (and disobey) the island rule. *Proceedings of the National Academy of Sciences of the United States of America* 116: 17632–17634.
- Blomberg SP, Garland T. 2002. Tempo and mode in evolution: Phylogenetic inertia, adaptation and comparative methods. *Journal of Evolutionary Biology* 15: 899–910.
- Blomberg SP, Garland Jr. T, Ives AR. 2003. Testing for phylogenetic signal in comparative data: Behavioral traits are more labile. *Evolution* 57: 717–745.
- Bowen L, Van Vuren D. 1997. Insular endemic plants lack defenses against herbivores. *Conservation Biology* 11: 1249–1254.
- Burns K. 2016. Size changes in island plants: Independent trait evolution in *Alyxia ruscifolia* (Apocynaceae) on Lord Howe Island. *Biological Journal of the Linnean Society* 119: 847–855.
- Burns K. 2019. Evolution in isolation: The search for an island syndrome in plants. In: Burns KC ed. *Evolution in isolation: The search for an island syndrome in plants*. Cambridge: Cambridge University Press. i–ii.
- Carlquist S. 1980. *Hawaii: A natural history. Geology, climate, native flora and fauna above the shoreline*. Honolulu: SB Printers Inc. for Pacific Tropical Botanical Garden (xii), 468p-illus., col. illus., maps. En Icones, Maps. Geog.
- Carlquist SJ. 1974. *Island biology*. New York: Columbia University Press.
- Carr GD. 1987. Beggar's ticks and tarweeds: Masters of adaptive radiation. *Trends in Ecology & Evolution* 2: 192–195.
- Carr GD, Kyhos DW. 1981. Adaptive radiation in the Hawaiian Silversword Alliance (Compositae-Madiinae). I. Cytogenetics of spontaneous hybrids. *Evolution* 35: 543–556.
- Caujapé-Castells J, Tye A, Crawford DJ, Santos-Guerra A, Sakai A, Beaver K, Lobin W, Florens FV, Moura M, Jardim R. 2010. Conservation of oceanic island floras: Present and future global challenges. *Perspectives in Plant Ecology, Evolution and Systematics* 12: 107–129.
- Cavaleri MA, Ostertag R, Cordell S, Sack L. 2014. Native trees show conservative water use relative to invasive trees: Results from a removal experiment in a Hawaiian wet forest. *Conservation Physiology* 2: cou016.
- Cordell S, Cabin RJ, Hadway LJ. 2002. Physiological ecology of native and alien dry forest shrubs in Hawaii. *Biological Invasions* 4: 387–396.
- Craven D, Knight TM, Barton KE, Bialic-Murphy L, Cordell S, Giardina CP, Gillespie TW, Ostertag R, Sack L, Chase JM. 2018. Opennahale: The open Hawaiian forest plot database. *Biodiversity Data Journal* 6: e28406.
- Denslow JS. 2003. Weeds in paradise: Thoughts on the invasibility of tropical islands. *Annals of the Missouri Botanical Garden* 90: 119–127.
- Díaz S, Kattge J, Cornelissen JH, Wright IJ, Lavorel S, Dray S, Reu B, Kleyer M, Wirth C, Prentice IC. 2022. The global spectrum of plant form and function: Enhanced species-level trait dataset. *Scientific Data* 9: 755.
- Frazier AG, Giambelluca TW. 2017. Spatial trend analysis of Hawaiian rainfall from 1920 to 2012. *International Journal of Climatology* 37: 2522–2531.
- Funk JL. 2013. The physiology of invasive plants in low-resource environments. *Conservation Physiology* 1: coto26.
- Funk JL, Throop HL. 2010. Enemy release and plant invasion: Patterns of defensive traits and leaf damage in Hawaii. *Oecologia* 162: 815–823.
- Gardner RC. 1976. Evolution and adaptive radiation in *Lipochaeta* (Compositae) of the Hawaiian Islands. *Systematic Botany* 1: 383–391.
- Garland Jr. T, Dickerman AW, Janis CM, Jones JA. 1993. Phylogenetic analysis of covariance by computer simulation. *Systematic Biology* 42: 265–292.
- Gillespie RG, Baldwin BG. 2010. Island biogeography of remote archipelagoes. In: Losos JB, Ricklefs RE eds. *The theory of island biogeography revisited*. Princeton and Oxford: Princeton University Press. 358–387.
- Givnish TJ, Millam KC, Mast AR, Paterson TB, Theim TJ, Hipp AL, Henss JM, Smith JF, Wood KR, Sytsma KJ. 2009. Origin, adaptive radiation and diversification of the Hawaiian lobeliads (Asterales: Campanulaceae). *Proceedings of the Royal Society B: Biological Sciences* 276: 407–416.
- Givnish TJ, Montgomery RA. 2014. Common-garden studies on adaptive radiation of photosynthetic physiology among Hawaiian lobeliads. *Proceedings of the Royal Society B: Biological Sciences* 281: 20132944.
- Heberling JM, Fridley JD. 2012. Biogeographic constraints on the world-wide leaf economics spectrum. *Global Ecology and Biogeography* 21: 1137–1146.
- Heberling JM, Mason NW. 2018. Are endemics functionally distinct? Leaf traits of native and exotic woody species in a New Zealand forest. *PLoS One* 13: e0196746.

- Hulme PE, Bernard-Verdier M. 2018. Evaluating differences in the shape of native and alien plant trait distributions will bring new insights into invasions of plant communities. *Journal of Vegetation Science* 29: 348–355.
- Kagawa A, Sack L, Duarte Ke, James S. 2009. Hawaiian native forest conserves water relative to timber plantation: Species and stand traits influence water use. *Ecological Applications* 19: 1429–1443.
- Kattge J, Bönisch G, Díaz S, Lavorel S, Prentice IC, Leadley P, Tautenhahn S, Werner GD, Aakala T, Abedi M. 2020. Try plant trait database—Enhanced coverage and open access. *Global Change Biology* 26: 119–188.
- König C, Weigelt P, Taylor A, Stein A, Dawson W, Essl F, Pergl J, Pyšek P, van Kleunen M, Winter M. 2021. Source pools and disharmony of the world's island floras. *Ecography* 44: 44–55.
- Kores P. 1980. Pollination mechanisms as a limiting factor in the development of the orchidaceous flora of Hawai'i. Department of Botany, University of Hawaii at Manoa.
- Leishman MR, Haslehurst T, Ares A, Baruch Z. 2007. Leaf trait relationships of native and invasive plants: Community- and global-scale comparisons. *New Phytologist* 176: 635–643.
- Leitão RP, Zuanon J, Villéger S, Williams SE, Baraloto C, Fortunel C, Mendonça FP, Moullot D. 2016. Rare species contribute disproportionately to the functional structure of species assemblages. *Proceedings of the Royal Society B: Biological Sciences* 283: 20160084.
- Mathakutha R, Steyn C, le Roux PC, Blom IJ, Chown SL, Daru BH, Ripley BS, Louw A, Greve M. 2019. Invasive species differ in key functional traits from native and non-invasive alien plant species. *Journal of Vegetation Science* 30: 994–1006.
- Mologni F, Bellingham PJ, Cameron EK, Dinh K, Wright AE, Burns KC. 2022. Functional traits explain non-native plant species richness and occupancy on northern New Zealand islands. *Biological Invasions* 24: 2135–2154.
- Montesinos D. 2022. Fast invasives fastly become faster: Invasive plants align largely with the fast side of the plant economics spectrum. *Journal of Ecology* 110: 1010–1014.
- Münkemüller T, Lavergne S, Bzeznik B, Dray S, Jombart T, Schiffrers K, Thuiller W. 2012. How to measure and test phylogenetic signal. *Methods in Ecology and Evolution* 3: 743–756.
- Murakami H, Wang B, Li T, Kitoh A. 2013. Projected increase in tropical cyclones near Hawaii. *Nature Climate Change* 3: 749–754.
- Ottaviani G, Keppel G, Götzenberger L, Harrison S, Opedal ØH, Conti L, Liancourt P, Klimešová J, Silveira FAO, Jiménez-Alfaro B, Negoita L, Doležal J, Hájek M, Ibanez T, Méndez-Castro FE, Chytrý M. 2020. Linking plant functional ecology to island biogeography. *Trends in Plant Science* 25: 329–339.
- Pattison R, Goldstein G, Ares A. 1998. Growth, biomass allocation and photosynthesis of invasive and native Hawaiian rainforest species. *Oecologia* 117: 449–459.
- Peñuelas J, Sardans J, Llusia J, Owen SM, Carnicer J, Giambelluca TW, Rezende EL, Waite M, Niinemets Ü. 2010. Faster returns on 'leaf economics' and different biogeochemical niche in invasive compared with native plant species. *Global Change Biology* 16: 2171–2185.
- Perez-Harguindeguy N, Díaz S, Garnier E, Lavorel S, Poorter H, Jaureguiberry P, Bret-Harte MS, Cornwell WK, Craine JM, Gurvich DE. 2016. Corrigendum to: New handbook for standardised measurement of plant functional traits worldwide. *Australian Journal of Botany* 64: 715–716.
- Price JP, Wagner WL. 2004. Speciation in Hawaiian angiosperm lineages: Cause, consequence, and mode. *Evolution* 58: 2185–2200.
- Price JP, Wagner WL. 2018. Origins of the Hawaiian flora: Phylogenies and biogeography reveal patterns of long-distance dispersal. *Journal of Systematics and Evolution* 56: 600–620.
- R Core Team. 2021. R: A language and environment for statistical computing. Vienna: R Foundation for Statistical Computing.
- Reich PB. 2014. The world-wide 'fast-slow' plant economics spectrum: A traits manifesto. *Journal of Ecology* 102: 275–301.
- Revell LJ. 2009. Size-correction and principal components for interspecific comparative studies. *Evolution: International Journal of Organic Evolution* 63: 3258–3268.
- Revell LJ. 2012. Phytools: An R package for phylogenetic comparative biology (and other things). *Methods in Ecology and Evolution* 3: 217–223.
- Revell LJ, Harmon LJ. 2008. Testing quantitative genetic hypotheses about the evolutionary rate matrix for continuous characters. *Evolutionary Ecology Research* 10: 311–331.
- Santiago LS, Goldstein G, Meinzer FC, Fownes JH, Mueller-Dombois D. 2000. Transpiration and forest structure in relation to soil waterlogging in a Hawaiian montane cloud forest. *Tree Physiology* 20: 673–681.
- Schrader J, Wright IJ, Kreft H, Westoby M. 2021. A roadmap to plant functional island biogeography. *Biological Reviews* 96: 2851–2870.
- Simberloff D. 1995. Why do introduced species appear to devastate islands more than mainland areas? *Pacific Science* 49: 87–97.
- Smith SA, Brown JW. 2018. Constructing a broadly inclusive seed plant phylogeny. *American Journal of Botany* 105: 302–314.
- Sperry JH, O'Hearn D, Drake DR, Hruska AM, Case SB, Vizentin-Bugoni J, Arnett C, Chambers T, Tarwater CE. 2021. Fruit and seed traits of native and invasive plant species in Hawai'i: Implications for seed dispersal by non-native birds. *Biological Invasions* 23: 1819–1835.
- Stone BC. 1966. *Studies in the Hawaiian Rutaceae: A conspectus of species and varieties and some further new taxa in the genus Pelea A. Gray*. VII. Honolulu: Bernice P. Bishop Museum.
- Stratton L, Goldstein G. 2001. Carbon uptake, growth and resource-use efficiency in one invasive and six native Hawaiian dry forest tree species. *Tree Physiology* 21: 1327–1334.
- Timm OE, Giambelluca TW, Diaz HF. 2015. Statistical downscaling of rainfall changes in Hawai'i based on the CMIP5 global model projections. *Journal of Geophysical Research: Atmospheres* 120: 92–112.
- Trauernicht C, Pickett E, Giardina CP, Litton CM, Cordell S, Beavers A. 2015. The contemporary scale and context of wildfire in Hawai'i. *Pacific Science* 69: 427–444.
- van Kleunen M, Weber E, Fischer M. 2010. A meta-analysis of trait differences between invasive and non-invasive plant species. *Ecology Letters* 13: 235–245.
- Wagner W, Herbst D, Lorence D. 2017. Flora of the Hawaiian Islands website. <https://naturalhistory2.si.edu/botany/hawaiianflora/> [accessed January 2017].
- Wagner WL, Herbst DR, Sohmer SH. 1999. *Manual of the flowering plants of Hawai'i*. Honolulu: University of Hawai'i and Bishop Museum Press. 1 and 2.
- Wagner WL, Weller S, Sakai A. 1995. Phylogeny and biogeography in Schiedea and Alsindendron (Caryophyllaceae). In: Wagner WL, Funk VA eds. *Hawaiian biogeography: Evolution on a hot spot archipelago*. Washington D.C.: Smithsonian Institution Press. 221–258.
- Warton DI, Duursma RA, Falster DS, Taskinen S. 2012. Smatr 3—An R package for estimation and inference about allometric lines. *Methods in Ecology and Evolution* 3: 257–259.



- Warton DI, Wright IJ, Falster DS, Westoby M. 2006. Bivariate line-fitting methods for allometry. *Biological Reviews* 81: 259–291.
- Westerband AC, Knight TM, Barton KE. 2020. Intraspecific trait variation and reversals of trait strategies across key climate gradients in native Hawaiian plants and non-native invaders. *Annals of Botany* 127: 553–564.
- Wright IJ, Reich PB, Westoby M, Ackerly DD, Baruch Z, Bongers F, Cavender-Bares J, Chapin T, Cornelissen JHC, Diemer M, Flexas J, Garnier E, Groom PK, Gulias J, Hikosaka K, Lamont BB, Lee T, Lee W, Lusk C, Midgley JJ, Navas M-L, Niinemets Ü, Oleksyn J, Osada N, Poorter H, Poot P, Prior L, Pyankov VI, Roumet C, Thomas SC, Tjoelker MG, Veneklaas EJ, Villar R. 2004. The worldwide leaf economics spectrum. *Nature* 428: 821.
- Ziegler AC. 2002. *Hawaiian natural history, ecology, and evolution*. Honolulu: University of Hawaii Press.

## Supplementary Material

The following supplementary material is available online for this article at <http://onlinelibrary.wiley.com/doi/10.1111/jse.13032/supinfo>:

**Table S1.** Hawaiian species list. Plant species were sampled across the Hawaiian Islands in the present study. Endemic and indigenous were combined to form one group called “native” for all analyses ( $N = 52$ ). All species were broad-leaved and woody. \*Deciduous. †This species is among the 25 most abundant species in the Hawaiian archipelago, based on the OpenNahele plot database (Craven et al., 2018), which includes both native and nonnative species. In total, there were 405 endemic species, and 148 indigenous species, which includes one climber, three climber/shrub, 18 shrubs, nine shrub/tree, and 21 trees.

**Table S2.** Continental species. Continental species ( $N = 854$ ) were included in the present study, for which data were extracted from publicly available trait data sets. Woodiness, growth form, and leaf type were extracted from the “Global Spectrum of Plant Form and Function Database” (Díaz et al., 2022). Growth form could not be assigned in cases when the specific epithet was not available. Shading indicates cases for which one of the above three properties was not available and therefore filled in using external data sources. Information regarding deciduousness was omitted due to a lack of available data.

**Table S3.** Data sources. Trait data sets were used to compare native Hawaiian plants to continental species. All data sets were accessed via the TRY database. Number of species denotes the number of species that have data for all four focal traits. The data sets were combined, that is, the trait data were averaged across studies, to produce a grand total of 854 continental species. Note that 36 species were removed because they could not be placed on the phylogenetic tree and 53 species were removed from the Global  $A_{\max}$  data set because duplicate entries (with identical trait values) were contained within the Global Respiration data set.

**Fig. S1.** Phylogenetic tree for native Hawaiian species. Evolutionary relationships among native Hawaiian species are included in the present study.

**Fig. S2.** Phylogenetic tree for all study species. Phylogenetic relationships between the 906 species are included in the present study, including 52 native Hawaiian species (indicated with red arrows) and 854 non-Hawaiian continental species.

**Table S4.** Estimates of phylogenetic signal. Estimates of phylogenetic signal for four leaf traits in (A) native Hawaiian species and (B) all species included in the present study.  $\lambda$  values close to 1 indicate a strong phylogenetic signal, such that traits among related species are more similar than what would be expected under Brownian motion. Blomberg's  $K$  can take on values greater than 1.  $P$  values are based on 1000 randomizations. LMA, leaf mass per area ( $\text{g/m}^2$ ),  $A_{\text{mass}}$ , mass-based photosynthetic rate ( $\text{nmol CO}_2/\text{g per s}$ ),  $N_{\text{mass}}$ , mass-based leaf nitrogen concentration (%),  $P_{\text{mass}}$ , mass-based leaf phosphorus concentration (%).

**Table S5.** Phylogenetic analysis of variance (ANOVA). Simulation-based phylogenetic ANOVA, evaluating differences between native Hawaiian and a globally distributed sample of continental species. Statistical significance is indicated using boldface.

**Table S6.** Phylogenetic reduced major axis regression.  $H_0$  is the null hypothesis for each trait relationship, and the observed slope indicates the effect of phylogenetic relatedness. Significant effects are indicated in boldface.

**Table S7.** Test of equal slopes. Bivariate leaf trait relationships for native Hawaiian plants versus a globally distributed sample of continental species. The null hypothesis is that the slopes are equal, which was tested using reduced major axis regression on log-scaled leaf traits. LMA, leaf mass per area ( $\text{g/m}^2$ ),  $A_{\text{mass}}$ , mass-based photosynthetic rate ( $\text{nmol CO}_2/\text{g per s}$ ),  $N_{\text{mass}}$ , mass-based leaf nitrogen (%),  $P_{\text{mass}}$ , mass-based leaf phosphorus (%). Boldface indicates statistically significant slope differences.

**Table S8.** Test of equal slopes, excluding herbaceous species. Bivariate leaf trait relationships for native Hawaiian plants versus a globally distributed sample of continental species. We excluded all herbaceous species to reduce variation due to life forms. The null hypothesis is that the slopes are equal, which was tested using reduced major axis regression on log-scaled leaf traits. LMA, leaf mass per area ( $\text{g/m}^2$ ),  $A_{\text{mass}}$ , mass-based photosynthetic rate ( $\text{nmol CO}_2/\text{g per s}$ ),  $N_{\text{mass}}$ , mass-based leaf nitrogen (%),  $P_{\text{mass}}$ , mass-based leaf phosphorus (%). Boldface indicates statistically significant slope differences.

**Table S9.** Principal component loadings for phylogenetically controlled principal components analysis, using all species. Loadings of resource use traits onto each of the first two principal components axes.

**Fig. S3.** Phylogenetically controlled principal components analysis for native Hawaiian species alone. Trait space of native Hawaiian species using a phylogenetically controlled analysis. Each point represents a species.

**Table S10.** Principal component loadings for phylogenetically controlled principal components analysis, for only native Hawaiian species. Loadings of resource-use traits onto each of the first two principal components axes.

**Fig. S4.** Phylogenetically controlled principal components analysis excluding herbaceous species. Trait space of native Hawaiian species versus a sample of continental species, excluding herbaceous species. Compared with Fig. 2, note the lower density of gray points in the center of the figure. This graph was generated using a phylogenetically controlled analysis. Each point represents a species.